Fiscal Year:	FY 2023	**Task Last Updated:**	FY 10/26/2023
PI Name:	Ferri, James Ph.D.		
Project Title:	PASTA: Particle Stabilized Emulsions and Foams		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Previous affiliation was Lafayette College, Easton, PA until early 2018 when PI moved to Virginia Commonwealth University		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	06/01/2018	**End Date:**	08/29/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:	ISS		
Flight Assignment:	NOTE: Continuation grant awarded, 80NSSC21K2080, extending project to 8/29/2023 (Ed., 1/31/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K2080 ; 80NSSC18K0453		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

ED. NOTE January 2022: New grant awarded September 2021, 80NSSC21K2080, extending project to 8/29/2023. Continuation of "Particle Stabilized Emulsions and Foams-80NSSC18K0453" in cooperation with European team, Emulsion Dynamics and Drop Interfaces.
ED. NOTE July 2018: Continuation of "Particle Stabilized Emulsions and Foams," grant NNX15AE98G, due to Principal Investigator James Ferri's move to Virginia Commonwealth University (VCU) in spring 2018.

The overall aim of the PASTA (Particle Stabilized Emulsions and Foams) project is the development of a mechanistic description of the stabilization or destabilization of foams and emulsions, which can be accomplished by various combinations of surfactants, polymers, and particles. This project is continuing as EDDI (Emulsion Dynamics and Drop Interfaces).

Emulsions are liquid-liquid dispersions of significant technological importance. Emulsions consist of droplets of one phase dispersed in a matrix of a phase containing a different liquid. Emulsions are thermodynamically unstable, so that at equilibrium, these liquid phases separate. The existence of kinetically stable emulsions relies in the presence of emulsifiers. These are blends of surface-active species, such as short-chain surfactants, polymers, proteins, and even nanoparticles, which accumulate at the liquid-liquid interface and decrease the energy needed for the creation of new interfaces. These species modify the physicochemical and mechanical features of the interface and counteract destabilization. In contrast, de-emulsifiers can be used to counteract the stabilizing effects present in some emulsion systems (e.g., natural emulsions) to destabilize and separate the liquid phases. These processes are referred to as emulsion breaking. However, a general mechanism of the role of surface-active species in the stabilization of an emulsion has yet to be elucidated. Therefore the development of emulsifier or de-emulsifier formulations are still largely empirical.

The EDDI project, the dynamic behavior and evolution of emulsions will be investigated, including efforts to characterize a) single droplets and b) multi-droplet systems.

a) Single droplets:

•Characterization of the chemico-physical and mechanical features of liquid-liquid interfaces, such as the adsorption dynamics and the dilational rheology, measuring the concerned quantities (dynamic surface tension, dilational viscoelasticity, etc.) on a large dynamic range.

•Develop models for the above features, accounting for the dynamic and structural aspects of adsorption layers. Depending on the system, surfactant exchange between the liquid phases may occur, which also need to be investigated and incorporated in the models.

b) Droplet-droplet & droplet-bubble interactions:

•Investigate and define the conditions for coalescence (drop-drop) and aggregation/incorporation (drop-drop, drop-bubbles) between single droplets, measuring the corresponding rates.

•Investigate the Ostwald Ripening for doublets of droplets, measuring the corresponding rates.

•Correlate the results of these investigations with single drop dynamics.

In addition to the experimental program described above, theory will be developed which describes the behavior above in terms of the molecular and intrinsic constants associated with the materials of which they are composed.

The composition of the consortium of participating teams (USA, Germany, Italy, Greece) is such that all aspects of the formulated problem are covered. The teams demonstrate a complementary expertise in form of the scientific fields and available experimental tools, as well as capacity for developing and applying the relevant theoretical frameworks.

Key experiments in microgravity conditions support the specific aims of this project. The basic processes involved in emulsions and foams, such as destabilization, mainly droplet aggregation and coalescence (destabilization of the film between droplets/bubbles), will be studied in the absence of segregation for model emulsions. In fact, during emulsion and foam destabilization, droplets larger than ten microns are rapidly segregating, making impossible on Earth the investigation of the above processes in controlled conditions. Microgravity experiments will then provide information about the features and characteristic times of these processes to be utilized to parameterize and validate the respective models. Since its inception, the PASTA project has sought to leverage the existing facilities on International Space Station (ISS) Fluid Science Laboratory (FSL). During the past few years, ESA (the European Space Agency) has refocused support in the Soft Matter Dynamics thrust to emphasize the Foam Optics And Mechanics-Coarsening (FOAM-C) facility which was originally developed to study coarsening and stability, respectively, of wet foams (multiple, polydisperse interfaces). As a result of shift in ESA priorities, in 2017 the PASTA team piloted the existing experimental module, FOAM-C, for suitability for emulsion dynamics studies.

DWS (diffusing wave spectroscopy) has been implemented as the main diagnostics of the ESA Soft Matter Dynamics (SMD, formerly FoamC) facility, which has been uploaded onboard the ISS in July 2018, to perform a first set of experiments on granular materials and foams. The presence of the line-camera on the SMD facility will allow the so-called multispeckle analysis, i.e., averaging the evolution of different speckles of scattered light at the same time. With respect to the simple time average done with data from single detectors, multispeckle offers the advantage to reach good statistics in short time, besides enabling to compute time resolved correlation functions, two-time correlation functions, and higher order correlation functions, which are useful to investigate transient states and residual dynamics in arrested systems.

In the framework of the present project, these approaches will be used to follow the temporal evolution of droplet sizes and their dynamics, which is associated to the mechanical properties of aging emulsions, with particular emphasis on low surfactant concentrations.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	Emulsions have large relevance for applications in a wide spectrum of technologies, processes, and products in different fields including oil and gas, specialty materials, pharmaceuticals, cosmetics, and foods. Both stable and unstable emulsions are needs that are shared across a variety of applications. For example, efficient de-emulsifiers are needed to facilitate the process of phase separation (breaking) in the crude oil emulsions extracted from the wells, while high stability oil-in-water emulsions are required in the final stage of production of the same well or to facilitate pumping high-viscosity oils through pipelines. In the food industry, many products based on emulsified fats or oils (condiments, sauces, frozen confections, and others) require long term stability. However, destabilizing emulsions are key to processing raw food materials such as milk and vegetable oils.
Task Progress:	A suite of analytical technologies on the International Space Station was utilized for the development of a mechanistic description of stabilization or destabilization of foams and emulsions under microgravity conditions. Emulsions are widely used in natural and synthetic products, technologies, and industrial processes. Inherently, emulsions are thermodynamically unstable, and the kinetic stability observed is promoted by the surfactants and/or other additives. To date, emulsion technology relies mostly on empirical approaches. There are many objectives related to emulsions in applications in foods, pharmaceuticals, and the oil industry. A few valuable targets include identification of effective and resource-saving/milder emulsifiers and de-emulsifying agents to reduce environmental footprint and adverse health effects, control of the droplet size in emulsified products, and increased shelf-life and stability of emulsified products. It is necessary to understand the link between emulsions features and the physical chemistry of droplet interfaces to utilize by design formulation of emulsions. Understanding the effects of surfactants and other additives on single destabilization mechanisms driving the behavior during emulsion aging is imperative. This type of mechanistic knowledge is possible by using a multi-scale investigation strategy from single interfaces to droplet evolution and dynamics, and extending studies to be done in microgravity conditions allows for even more controlled isolation of droplet dynamics and mechanisms. *The project elucidated and modeled the relations between the chemico-physical properties of the droplet interface and the features of the mechanisms responsible for emulsion destabilization, such as droplet coalescence and Ostwald ripening. *Benchmark experiments have been executed for 16 formulations, on board the International Space Station (ISS), exploiting the weightlessness (microgravity) conditions to investigate the above mechanisms as decoupled from terrestrial creaming. These studies concerned model emulsions formed between water and Medium Chain Triglyceride (MCT) oil, stabilized by non-ionic polyoxyethylenated alcohol surfactants. *The experiments on board the ISS utilized the European Space Agency (ESA) facility Soft Matter Dynamics (SMD) formerly referred to as the FOAM-C, and utilized a unique methodology/analysis developed using Diffusing Wave Spectroscopy (DWS) as an effective diagnostic tool to investigate the emulsion aging and the droplet's dynamic regimes. The sensitivity of the DWS detector is sufficient to measure an evolution of the emulsion, and the data analysis was optimized to be effective for a quantitative analysis of the mechanistic phenomena. *Qualitative and quantitative insight related to the phenomena associated with aging emulsions in conditions that are not reproducible on ground were uncovered. * Some challenges remain regarding the filling of the cell, with respect to the presence of air bubbles. Residual air has sizable effects, as seen by fluctuations in the earlier stages, while at longer times, it is likely that air accumulates in large bubbles in sample cell corners, resulting in fewer fluctuations. Since project inception, a number of publications have been submitted by the Virginia Commonwealth University (VCU) team and collaborators. [Ed. Note: See Bibliography.]
Bibliography Type:	Description: (Last Updated: 12/03/2024)
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